

Krishnaa Vadivel | Software Engineer | Machine Learning | Distributed Computing

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Summary

Software engineer and Yale PhD researcher who developed machine-learning models, numerical software, and distributed applications in Python and C++. Experience spans CUDA/ROCm acceleration, national-lab HPC systems, Linux-based development, and agentic software automation using modern LLM APIs and local models.

Experience

Yale University, Electrical Engineering

PhD Research Assistant

2021–present

- Developed graph neural networks and automatic-differentiation models in Python to predict material properties including optical emission spectra.
- Developed Python and C++ software for first-principles and many-body simulation using MPI, OpenMP, PETSc, SLEPc, CUDA, ROCm, and CMake across leadership-class HPC systems.
- Built distributed applications for materials and molecular simulations on national-lab GPU clusters including Frontier, Frontera, and Perlmutter.
- Worked across research code, debugging, compiler-aware builds, documentation, and theory-to-implementation translation for scalable computational software.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and presented APS talks in 2024, 2025, and 2026, demonstrating research output alongside production-style software work.

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Built parallel applications with CUDA and OpenCL fallback for large sensor-data and waveform processing, diagnostic analytics, and field-data visualization.
- Developed full-stack server applications with ASP.NET and Microsoft SQL Server for real-time control of company tool testing, data collection, and data processing.
- Developed C# GUI applications for preparing customer inputs and interacting with deployed hardware systems.

FindLight

Photonics Technical Writer

2018–2019

- Wrote technical content translating specialized optics and photonics topics into clear engineering-facing explanations.

Selected Projects

Machine Learning Models for Material Properties: Developed PyTorch Geometric graph neural networks and automatic-differentiation models for molecular-property prediction and optical-properties calculations in material systems.

Agentic Coding and Scientific Automation: Built LangChain-based workflows using OpenAI API models and local Ollama models to read documentation, generate structured inputs, and accelerate coding and research workflows.

Container Orchestration for First-Principles Computing: Developed CUDA-ready and cross-environment containers for first-principles software stacks and published them to Docker Hub for portable scientific computing.

Matrix-Multiply AI Accelerator: Developed a Verilog-based matrix-multiply accelerator project for AI-oriented compute pipelines and hardware-software performance exploration.

Education

Yale University

PhD, Electrical Engineering

2021–present

Yale University

MPhil, Electrical Engineering

2023

Yale University

MS, Electrical Engineering

2023

Cornell University

MEng, Electrical and Computer Engineering

2017–2018

Cornell University

BS, Electrical and Computer Engineering

2013–2017

Technical Skills

Languages: Python, C, C++, JavaScript, Bash, SQL

Scientific Computing: MPI, OpenMP, PETSc, SLEPc, NumPy, pandas, SciPy, SymPy, matplotlib

Systems: Linux, CMake, Docker, Docker Compose, Kubernetes, gcloud

Platforms: CUDA, ROCm, Frontier, Frontera, Perlmutter

ML / Automation: PyTorch, PyTorch Geometric, scikit-learn, LangChain, OpenAI API, Ollama, graph neural networks, transformers, agentic automation